MLP / CIFAR-10 | three paths to 0.75x base LR | seed 0
Each intervention versus unchanged baseline; same Spark; solid unsmoothed curves

Cosine32 down to 0.75x
Hold from step 544

Cosine32 down to 0.5x
Abrupt rise to 0.75x at 650

Cosine32 down to 0.5x
Cosine70 rise to 0.75x over 650-720

Training loss
Unchanged baseline LR
Intervention
Raw gradient ρ_t
Raw gradient norm $\|g_t\|$
$\eta\hat{\kappa}_t$
Applied LR / base LR
Completed optimizer updates